INORGANIC CUMULATIVE EXAMINATION
September 15, 1990

Fluxtional Molecules

There are several "classical" types of fluxtionality (non-rigidity) shown by inorganic molecules. Show your understanding of these types by answering <u>four of the following five</u> questions in as much detail as time permits.

1. The ammonia molecule exhibits a simple and somewhat unique type of non-rigidity. Describe it, and classify it as having a relatively high, medium, or low activation energy.

2. The PF_5 and $Fe(CO)_5$ molecules are famous for non-rigidity. Describe the structural features of this non-rigidity in as much detail as possible. NMR is often used in these studies. Describe the spectra that are presented as part of this story for either PF_5 or $Fe(CO)_5$.

3. Chelated complexes such as $Ni(en)_3^{+2}$ interconvert from one optical isomer to the other. Describe the reaction intermediate proposed in the most common "internal twist" mechanism.

4. The 18-electron complex $Ru(\eta^4\text{-}C_8H_8)(CO)_3$ displays four peaks in the proton NMR at -115° and one peak at 25°. Explain.

5. The terminal and bridging CO groups in $Rh_4(CO)_{12}$ exchange positions. Describe a likely mechanism, and propose a spectral test.